📝 **Report: Stock Price Forecasting – AAPL Case Study**
Task Set-B | DataSynthis Internship Submission

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1. Introduction

This task explored **time-series forecasting of stock prices** using a mix of statistical and machine learning approaches. While the primary dataset used was **Apple Inc. (AAPL)** daily prices (2010–2024), the pipeline is fully generalizable to other equities.

Key objectives:

- Implement **baseline and statistical models** (Naïve, SES, ARIMA)
- Implement **machine learning models** (LSTM; Prophet prepared but not finalized due to time)
- Perform **static vs. rolling window evaluation**
- Generate **visualizations and comparative metrics**
- Deploy final solution to **Hugging Face Space** for demonstration

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2. Data & Feature Engineering

- Data: **AAPL (target) + SPY (benchmark)**, 2010–2024
- Engineered **39 features**, later pruned to 20–25 based on correlation, VIF, and redundancy.
- Categories: Lags, Moving Averages, RSI, MACD, Volatility (20d, 50d), Volume ratios, Market correlation, Time features (sin/cos encodings).
- Final dataset: **3,572 rows, 59 columns**

Key engineered features:

- **Returns:** log_return, target_return (next-day), spy_return
- **Volatility:** hl_range (strongest predictor, corr ~0.55), rolling_std (20d/50d)
- **Momentum:** RSI(14), MACD
- **Market factors:** rolling_beta, market_corr_30d

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3. Models Implemented

- **Baseline Models:**
  - Naïve Forecast (last value persists)
  - Simple Exponential Smoothing (SES)
- **Statistical Model:**
  - ARIMA (auto-parameterized, AIC/BIC selection)
- **Machine Learning Model:**
  - LSTM (2 hidden layers, dropout regularization, trained with RMSE loss)
  - Prophet prepared, not included in final comparison due to time

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4. Evaluation Design

Two strategies applied:

1. **Static Evaluation** – train on historical (2010–2022), predict test set (2023–2024)
2. **Rolling Window Evaluation** – expanding retraining window; simulates live deployment

Metrics tracked:

- RMSE (error magnitude)
- MAE (absolute deviation)
- R^2 (fit quality)
- Directional Accuracy (trend correctness, % > 50 is meaningful)

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5. Results

📊 **Static vs Rolling Window (Returns-Level)**

- **RMSE:** Lowest for LSTM (0.0144 static), ARIMA competitive in rolling (0.0177).
- **MAE:** LSTM ~0.0100, ARIMA ~0.0104 → nearly tied.
- **R^2:** Naïve collapses (<-0.9), LSTM slightly positive.
- **Directional Accuracy:** All models ~50–53%. SES best static (53.1%), LSTM best rolling (51.9%).

📉 **Observation:** Rolling windows introduce ~22–25% degradation in error for all models.

📈 **Price Prediction Accuracy**

- **Static Evaluation (2023–2024):**
    - Naïve: MAE ≈ $5.05, RMSE ≈ $6.27 → surprisingly strong baseline
    - SES: MAE ≈ $10.3
    - LSTM: MAE ≈ $17.9 (final error ≈ 25.7%)
    - ARIMA: MAE ≈ $27.9 (final error ≈ 22.3%)
- **Rolling Evaluation (2022–2024):**
    - SES outperformed others with lowest MAE ($6.0) and lowest final error (0.9%)
    - Naïve worsened (MAE $13.3)
    - ARIMA collapsed under rolling (MAE $81.3, RMSE $86.9)
    - LSTM degraded (MAE $33.7, but final error better than static: 9.4%)

🔑 **Key Takeaways**

- **Best static model: LSTM** (lowest RMSE, generalizes slightly better than ARIMA/SES)
- **Best rolling model: SES** (robust to distribution drift, small final error %)
- **Naïve baseline:** competitive, surprisingly resilient for short horizons
- **ARIMA:** fitted static data well but **broke under rolling evaluation**
- **LSTM:** struggles with price-level prediction drift but maintains trend accuracy

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6. Visual Insights (Selected Plots)

- **Feature Engineering:** Correlation heatmaps & importance plots show **hl_range** dominant (~0.55 corr).
- **Error Distributions:** ARIMA & LSTM had tighter, more symmetric errors; SES skewed slightly.
- **Static vs Rolling:** All models degrade ~20–25% in RMSE under rolling.
- **Price Predictions:** SES tracked actual prices better in rolling mode; ARIMA diverged.
- **Training Diagnostics:** LSTM loss stabilized fast, but validation MAE plateaued.

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7. Recommendations

- For **short-term rolling predictions**, **SES is the most stable and reliable**.
- For **longer horizons (static evaluation)**, **LSTM generalizes slightly better**, though sensitive to drift.
- A **hybrid ensemble (SES + LSTM)** may balance stability with adaptability.
- Future extensions:
    - Incorporate **Prophet** for seasonality
    - Test **Transformer-based models** (Temporal Fusion Transformer, Informer)
    - Include **market/macro features** (VIX, interest rates, sector ETFs)

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8. Deployment

- Hugging Face Space: **DataSynthis_ML_JobTask**
- Deployed with **Gradio app (app.py)**, allowing interactive selection of model and forecast horizon.

- Repository includes:
  - Notebook with end-to-end pipeline
  - Engineered dataset & feature list
  - Visuals (comparisons, error distributions, correlations)
  - README with setup instructions

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9. Conclusion

This project demonstrated that **simple statistical models (SES)** can outperform complex models under **rolling/live conditions**, while **deep learning (LSTM)** provides an edge in **static backtesting**. The results underline a key principle in time-series forecasting: **robustness and stability often matter more than complexity** in real-world deployment.

✅ **Deliverables Completed:**

- Clean documented **Notebook**
- Full **Feature Engineering & Selection** pipeline
- **Baseline, ARIMA, LSTM** implemented
- **Static & Rolling evaluation metrics** with tables + plots
- **Report (this document)**
- **Deployment on Hugging Face** (DataSynthis_ML_JobTask)